



HOW TO BREAK ENIGMA CODEC

ALAN MATHISON TURING

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
THE DEGREE OF BACHELOR OF SCIENCE (COMPUTER SCIENCE)
SCHOOL OF INFORMATION TECHNOLOGY
KING MONGKUT'S UNIVERSITY OF TECHNOLOGY THONBURI

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Academic Year 2024

Thesis Committee

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Abstract

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Keywords : Enigma / Encryption / Cryptanalytic

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List of Symbols

θ	Theta
d	distance
kg	Kilogram
m ²	Square Metre

Vocabularies and Abbreviations

DoF	Degree of Freedom
CoM	Center of Mass
KMUTT	King Mongkut's University of Technology Thonburi

Chapter 1

Introduction

This framework aims to design and build an integrated system that consists of a centralised data management platform in conjunction with a network of interconnected sensors for the verification of sensors and the identification of anomalies will proceed more quickly as a result of this assembly.

The system makes use of a wide variety of sensors, the most prominent of which are the water level and flow rate sensors. This design was inspired by the infrastructure of the water treatment testbed in the School of Computing at Newcastle University serving as its source of inspiration. These sensors send their readings to an aggregate server (AG), which is implemented as a Raspberry Pi in this case. The primary role of the AG is that of a data collector; after collecting the consolidated data, it is then sent on to the principal server. It is anticipated that, in a more comprehensive real-world scenario, additional aggregate servers will relay data to the primary server, which will ensure that field operations are carried out without any interruptions.

1.1 Background and Rationale

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1.2 Motivation

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1.3 Research Questions

The research questions addressed in this study are as follows:

1. Aggregate server create secure stream S by encoding with BCH coding which predetermine coding parameter to match with the central server
2. Aggregate server send reading in plain text along with secure stream to the central server

1.4 Objectives

The objectives of this research can be summarized as follows:

1. Aggregate server create secure stream S by encoding with BCH coding which predetermine coding parameter to match with the central server
2. Aggregate server send reading in plain text along with secure stream to the central server
3. central server decode secure stream with plain text. If match, the central server continue to process anomaly detection. If not, it will reject those reading and alert the user for unauthenticated readings.

1.5 Expected Benefits

According to the objective of this research, we expect two benefits as follow:

1. Aggregate server create secure stream S by encoding with BCH coding which predetermine coding parameter to match with the central server
2. Aggregate server send reading in plain text along with secure stream to the central server

1.6 Scope of Study

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1.7 Thesis Organization

This thesis is divided into five chapters as follows:

- Chapter 1 is the introduction of the study. This chapter describes the introduction of the study, explain more.
- Chapter 2 presents the literature review of this study. This chapter explains the relevance of models, theories and concepts, explain more.
- Chapter 3 proposes the research methodology. This chapter illustrates the research procedures by identifying the step and process of working, explain more.
- Chapter 4 describes the results and discussion. This chapter analyzes the data by using some proper statistical data analysis methods, explain more.
- Chapter 5 explains the conclusions and future works. This chapter concludes the thesis and future directions of the research, explain more.

Chapter 2

Literature Reviews

This framework aims to design and build an integrated system that consists of a centralised data management platform in conjunction with a network of interconnected sensors for the verification of sensors and the identification of anomalies will proceed more quickly as a result of this assembly.

2.1 Challenges It Faces Indicating The Need For This Research

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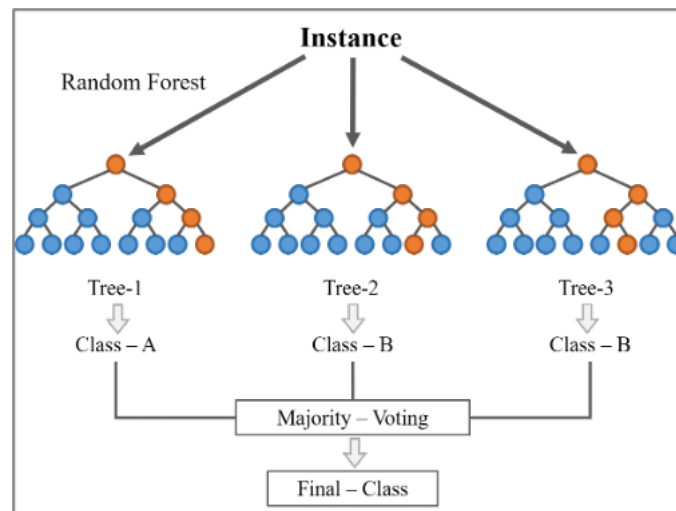


Figure 2.1: Random Forest Classification

From figure 2.1 shows an optimized model that endeavors to heighten detection rates, precision, and processing times [1]. The innovative architecture integrates the Catboost classifier, melding Gradient Boosting (GB) with Decision Trees (DT). This synergy serves to curtail gradient estimation bias, consequently amplifying the model's generalization prowess. A significant attribute of this model is its utilization of GPU during the training phase, ensuring swifter computations.

2.1.1 Fishbone Diagram

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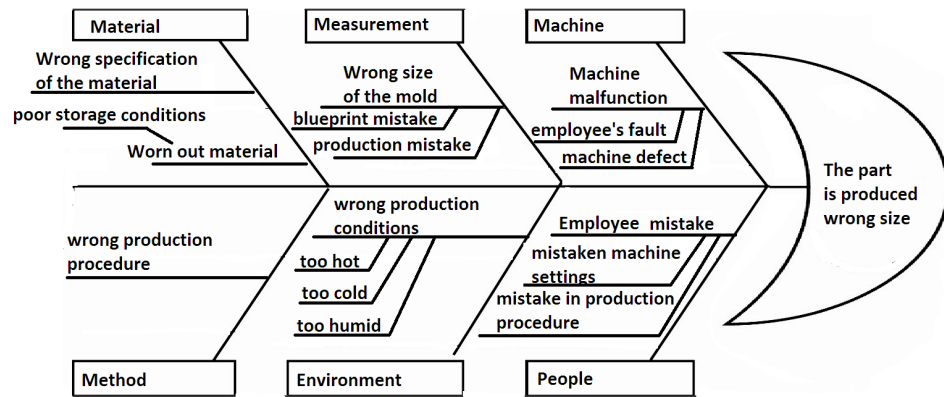


Figure 2.2: Fishbone Diagram

From figure 2.2 shows an optimized model that endeavors to heighten detection rates, precision, and processing times [1]. The innovative architecture integrates the Catboost classifier, melding Gradient Boosting (GB) with Decision Trees (DT). This synergy serves to curtail gradient estimation bias, consequently amplifying the model's generalization prowess. A significant attribute of this model is its utilization of GPU during the training phase, ensuring swifter computations.

2.1.2 Conclusion

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Chapter 3

Research Methodology

This framework aims to design and build an integrated system that consists of a centralised data management platform in conjunction with a network of interconnected sensors for the verification of sensors and the identification of anomalies will proceed more quickly as a result of this assembly.

3.1 RQ1: Your Research Question

3.1.1 Population and Sample

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2. Aggregate server send reading in plain text along with secure stream to the central server
3. central server decode secure stream with plain text. If match, the central server continue to process anomaly detection. If not, it will reject those reading and alert the user for unauthenticated readings.

3.2 RQ2: Your Research Question

3.2.1 Population and Sample

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3.3 RQ3: Your Research Question

3.3.1 Population and Sample

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Chapter 4

Results and Discussion

This framework aims to design and build an integrated system that consists of a centralised data management platform in conjunction with a network of interconnected sensors for the verification of sensors and the identification of anomalies will proceed more quickly as a result of this assembly.

4.1 Modeling Results

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Chapter 5

Conclusions

This framework aims to design and build an integrated system that consists of a centralised data management platform in conjunction with a network of interconnected sensors for the verification of sensors and the identification of anomalies will proceed more quickly as a result of this assembly.

5.1 Summary

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5.2 Contributions

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5.3 Future Works

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Appendix

Appendix A

Some Appendix

The contents...